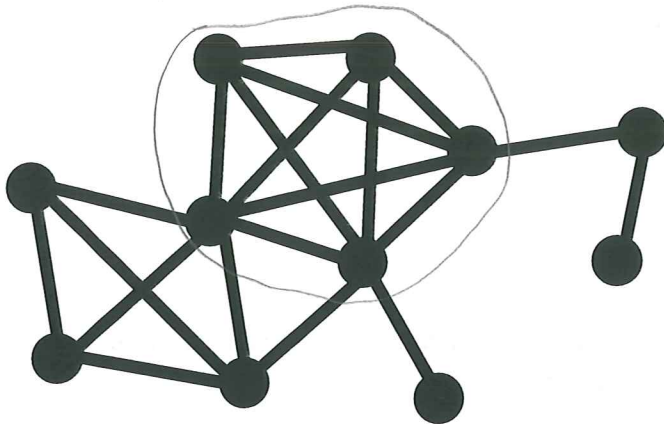


Name Brandon Buchanan

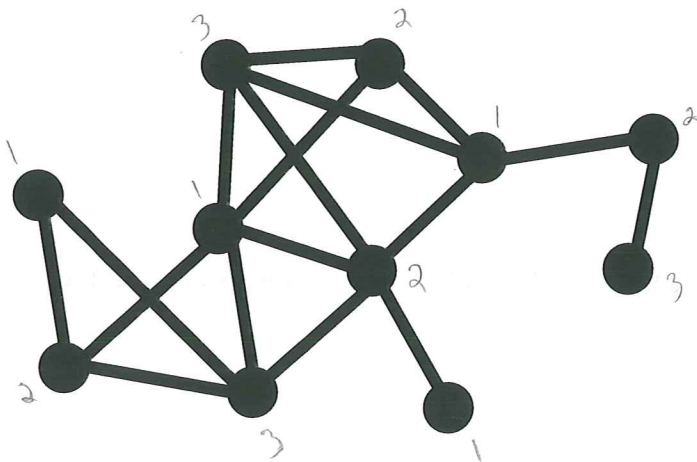
Find the largest Clique Size, i.e largest complete subgraph



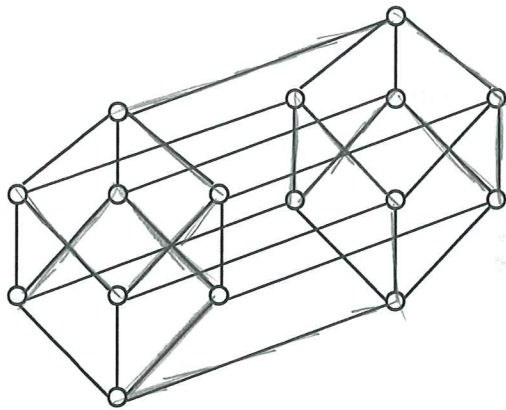
5

A coloring on a graph assigns each vertex a color and adjacent vertices can not have the same color.

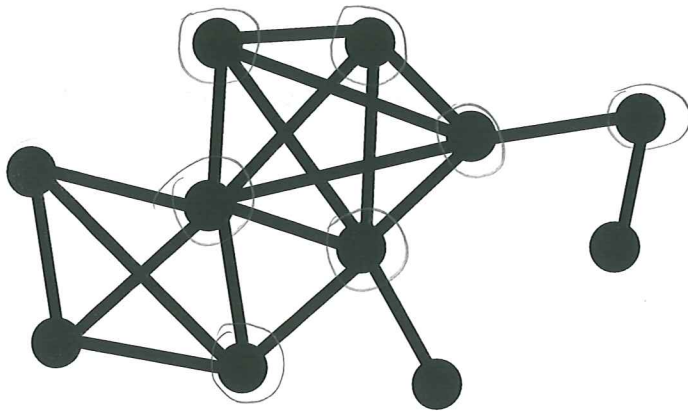
Can the graph be colored with 5 colors?
Find the chromatic number of the graph.



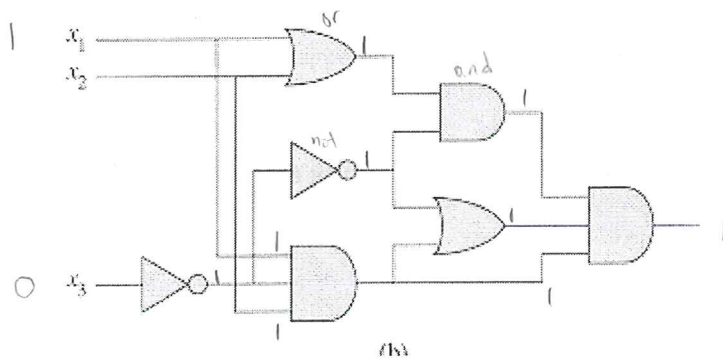
Does the following graph have a Hamiltonian cycle, a cycle including all the vertices?



Find the smallest vertex cover for the following graph, a vertex cover is a set of vertices U , where any edge of (u,v) of G must have either $u \in U$ or $v \in U$.



Circuit Satisfiability, can you assign values to x_1 , x_2 and x_3 which forces the output to 1



output is always 0

Suppose we are given an algorithm L which solves the longest path.

L : Input (Graph G , stating vertex v)

Output (Longest Path in G starting at v)

Use L to find whether G contains a Hamiltonian cycle in G , output Yes/No.